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ELECTRONIC DEVICE COMPRISING ELASTIC STRUCTURE SUPPORTING CONNECTING MEMBER

Abstract

An electronic device according to an example embodiment may include a housing including a plate part, a first circuit board disposed in the plate part adjacent to one side of the housing, which faces a first direction, a second circuit board disposed in the plate part adjacent to an opposite side of the housing, which faces a second direction being opposite to the first direction, a rear case coupled to the plate part to cover at least a portion of the second circuit board, an elastic structure coupled to the plate part adjacent to the second circuit board, and configured to provide an elastic force, and a connecting member connecting the first circuit board and the second circuit board, and a portion of which contacts the elastic structure. The rear case may include a first protruding part facing a portion of the elastic structure and a portion of the connecting member, which contacts the elastic structure. In addition, various embodiment recognized through the specification are possible.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of International Application No. PCT/KR2023/019347, filed on Nov. 28, 2023, in the Korean Intellectual Property Receiving Office, and claiming priority to Korean Patent Application No. 10-2022-0163168 filed Nov. 29, 2022, and Korean Patent Application No. 10-2023-0004004, filed on Jan. 11, 2023, the disclosures of which are all hereby incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] Certain example embodiments may relate to an electronic device including an elastic structure that supports a flexible printed circuit board (FPCB).

Background Art

[0003] In an electronic device, one or more printed circuit boards (PCBs) may be disposed in an interior of a housing. For example, the printed circuit boards may be disposed at an upper end and a lower end of the electronic device, respectively. The board disposed at the upper end and the board disposed at the lower end may be electrically connected to each other through a connecting member, such as a flexible printed circuit board (FPCB). The connecting member may be moved in the interior of the electronic device when the electronic device is moved or an impact is applied to the electronic device.

SUMMARY

[0004] An electronic device according to an example embodiment may include a housing including a plate part, a first circuit board disposed in the plate part adjacent to one side of the housing, which faces a first direction, a second circuit board disposed in the plate part adjacent to an opposite side of the housing, which faces a second direction being opposite to the first direction, a rear case coupled, directly or indirectly, to the plate part to cover at least a portion of the second circuit board, an elastic structure, comprising elastic material, coupled, directly or indirectly, to the plate part adjacent to the second circuit board, and configured to provide an elastic force, and a connecting member (e.g., a flexible circuit comprising conductive material and dielectric material) direct or indirectly electrically connecting the first circuit board and the second circuit board, and a portion of which contacts the elastic structure. The rear case may include a first protruding part, comprising a protrusion, facing a portion of the elastic structure and a portion of the connecting member, which contacts the elastic structure.

[0005] An electronic device according to an example embodiment may include a housing including a side member, wherein the side member may include a plate part and a frame part surrounding a periphery of the plate part, a first circuit board disposed in the plate part adjacent to one side of the housing, which faces a first direction, a second circuit board disposed in the plate part adjacent to an opposite side of the housing, which faces a second direction being opposite to the first direction, a rear case coupled, directly or indirectly, to the plate part to cover at least a portion of the second

circuit board, an elastic structure coupled, directly or indirectly, to the plate part adjacent to the second circuit board, and configured to provide an elastic force, and a connecting member connecting the first circuit board and the second circuit board, and a portion of which contacts the elastic structure. Movement of the connecting member may be guided in an interior of the housing based on elastic deformation of the elastic structure.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] FIG. 1 is a block diagram of an electronic device in a network environment, according to an example embodiment.

[0007] FIG. 2A is a front perspective view of an electronic device according to an example embodiment.

[0008] FIG. 2B is a rear perspective view of an electronic device according to an example embodiment.

[0009] FIG. 3 is an exploded perspective view of an electronic device according to an example embodiment.

[0010] FIG. 4A is a rear perspective view of an electronic device according to an example embodiment.

[0011] FIG. 4B is a rear perspective view of an electronic device according to an example embodiment.

[0012] FIG. 5 is a view illustrating an elastic structure of an electronic device according to an example embodiment.

[0013] FIG. 6 is a view illustrating a connecting member and an elastic structure of an electronic device according to an example embodiment.

[0014] FIG. 7 is a view illustrating driving of an elastic structure of an electronic device according to an example embodiment.

[0015] FIG. 8 is a view illustrating a rear case and a connecting member of an electronic device according to an example embodiment.

[0016] FIG. 9A is a cross-sectional view of an electronic device according to an example embodiment.

[0017] FIG. 9B is a cross-sectional view of an electronic device according to an example embodiment.

[0018] FIG. 10 is a view illustrating a rear case and a connecting member of an electronic device according to an example embodiment.

[0019] FIG. 11 is a cross-sectional view of an electronic device according to an example embodiment.

[0020] FIG. 12 is a view illustrating driving of an elastic structure and movement of a connecting member of an electronic device according to an example embodiment.

[0021] FIG. 13 is a view illustrating an operation of assembling an elastic structure, a second circuit board, a connecting member, and a second rear case of an electronic device according to an example embodiment.

[0022] FIG. 14 is a view illustrating a connecting member and an elastic structure of an electronic device according to an example embodiment.

[0023] FIG. 15 is a view illustrating an elastic structure of an electronic device according to an example embodiment.

[0024] FIG. 16 is a view illustrating a connecting member and an elastic structure of an electronic device according to an example embodiment.

[0025] FIG. 17 is a view illustrating an elastic structure of an electronic device according to an

example embodiment.

[0026] FIG. **18** is a view illustrating a connecting member and an elastic structure of an electronic device according to an example embodiment.

[0027] FIG. **19** is a view illustrating an elastic structure of an electronic device according to an example embodiment.

[0028] With regard to description of the drawings, the same or similar components will be denoted by the same or similar reference numerals.

DETAILED DESCRIPTION

[0029] Hereinafter, various embodiments of the disclosure will be described with reference to accompanying drawings. However, those of ordinary skill in the art will recognize that modification, equivalent, and/or alternative on various embodiments described herein can be variously made without departing from the scope and spirit of the disclosure.

[0030] FIG. **1** is a block diagram illustrating an electronic device **101** in a network environment **100** according to certain embodiments.

[0031] Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**.

[0032] According to an embodiment, the electronic device **101** may include a processor **120** comprising processing circuitry, a memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In certain example embodiments, at least one (e.g., the connecting terminal **178**) of the components may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In certain example embodiments, some of the components may be implemented as single integrated circuitry. For example, some (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) of the components may be implemented as embedded in the display module **160** (e.g., a display).

[0033] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled, directly or indirectly, with the processor **120**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **120** may load a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in a volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in a non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), and an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. When the electronic device **101** includes the main processor **121** and the auxiliary processor, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0034] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application).

According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., a neural network processing device) may include a hardware structure specified for processing an artificial intelligence (AI) model. The AI model may be generated through machine learning. The learning may be performed by the electronic device **101** performing the AI, and may be performed through an additional server (e.g., the server **108**). A learning algorithm may include, for example, a supervised learning algorithm, an unsupervised learning algorithm, a semi-supervised learning algorithm, or a reinforcement learning algorithm, but the disclosure is not limited thereto. The AI model may include a plurality of artificial neural network (ANN) layers. The ANN may include a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzman machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), a deep Q-networks or the combination of the above networks, but the disclosure is not limited thereto. The AI model may additionally or alternatively include a software structure, in addition to a hardware structure.

[0035] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0036] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0037] The input module **150** may receive a command or data to be used by other component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0038] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record, and the receiver may be used for an incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0039] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0040] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or an external electronic device (e.g., the electronic device **102**) (e.g., speaker of headphone) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0041] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0042] The interface **177** may support one or more specified protocols to be used for the electronic

device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0043] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0044] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0045] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0046] The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0047] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0048] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). The corresponding communication module from among the communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (WiFi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, 5G network, next generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0049] The wireless communication module **192** may support a 5G network and a next-generation communication technology, for example, a new radio (NR) access technology after a 4G network. The NR access technology may support high-speed transmission for high capacity data (enhanced mobile broadband; eMBB), terminal power minimizing and multiple terminal access (massive machine type communication; mMTC), or ultra-reliable and low-latency communications

(URLLC). The wireless communication module **192** may support a high-frequency band (e.g., mmWave band) to achieve, for example, a higher data rate. The wireless communication module **192** may support various technologies, for example, beamforming, massive multiple-input and multiple-output (MIMO), Full-dimensional MIMO, an array antenna, analog beam-forming, or a large-scale antenna, to secure performance in high frequency bands. The wireless communication module **192** may support various requirements defined in the electronic device **101**, the external electronic device (e.g., the electronic device **104**) or the network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for eMBB realization, loss coverage (e.g., 164 dB or less) for mMTC realization, or U-plane latency (e.g., 0.5 ms or less, or the round trip of 1 ms or less in each of a downlink (DL) and an uplink (UL)) for URLCC realization.

[0050] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., PCB). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., an array antenna). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0051] According to certain embodiments, the antenna module **197** may form an mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., a bottom surface) of the printed circuit board, or disposed adjacent to the first surface to support the specific high frequency band (e.g., mmWave band), and a plurality of antennas (e.g., an array antenna) disposed on a second surface (e.g., a top surface or a side surface) of the printed circuit board or disposed adjacent to the second surface to transmit or receive a signal having the specified high frequency band.

[0052] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0053] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, when the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide an ultra-latency service by using, for example,

distributed computing or mobile edge computing. According to certain embodiments, the external electronic device **104** may include the Internet of things (IoT). The server **108** may be an artificial server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to an artificial intelligence service (e.g., a smart home, a smart city, a smart car, or healthcare service) based on the 5G communication technology and the IoT-related technology.

[0054] FIG. 2A is a front perspective view of an electronic device according to an embodiment. FIG. 2B is a rear perspective view of an electronic device according to an embodiment.

[0055] Referring to FIGS. 2A and 2B, an electronic device **200** (e.g., the electronic device **101** of FIG. 1) according to an embodiment may include a housing **210** including a first surface (or a front surface) **210A**, a second surface (or a rear surface) **210B**, and a third surface (or a side surface) **210C** that surrounds a space between the first surface **210A** and the second surface **210B**. In various embodiments, the housing **210** may refer to a structure that defines a portion of the first surface **210A**, the second surface **210B**, and the third surface **210C**.

[0056] The first surface **210A** may be of or include a front plate **202** (e.g., a glass plate including various coating layers, or a polymer plate), at least a portion of which is substantially transparent. The second surface **210B** may be of or include a rear plate **211** that is substantially opaque. The rear plate **211**, for example, may be formed by coated or colored glass, ceramic, a polymer, a metal (e.g., aluminum, stainless steel (STS), or magnesium), or a combination of at least two of the materials. The third surface **210C** may be coupled, directly or indirectly, to the front plate **202** or the rear plate **211** and may be of or include a side bezel structure (or a side member) **218** including a metal and/or a polymer. In various embodiments, the rear plate **211** and the side bezel structure **218** may be integrally formed, and may include the same material (e.g., a metallic material, such as aluminum).

[0057] The front plate **202** may include two first areas **210D** that are bent from a partial area of the first surface **210A** toward the rear plate **211** and extend seamlessly. The first areas **210D** may be located at opposite ends of a long edge of the front plate **202**.

[0058] The rear plate **211** may include two second areas **210E** that are bent from a partial area of the second surface **210B** toward the front plate **202** and extend seamlessly. The second areas **210E** may be included at opposite ends of a long edge of the rear plate **211**.

[0059] In various embodiments, the front plate **202** (or the rear plate **211**) may include only one of the first areas **210D** (or the second areas **210E**). Furthermore, in various embodiments, the front plate **202** (or rear plate **211**) may not include some of the first areas **210D** (or the second areas **210E**).

[0060] The electronic device **200** may include at least one of a display **201** (e.g., the display module **160** of FIG. 1), an audio module **203**, **204**, and **207** (e.g., the audio module **170** of FIG. 1), a sensor module (not illustrated) (e.g., the sensor module **176** of FIG. 1), a camera module **205**, **212**, and **213** (e.g., the camera module **180** of FIG. 1), a key input device **217** (e.g., the input device **150** of FIG. 1), a light emitting element (not illustrated), and a connector hole **208** (e.g., the connection terminal **178** of FIG. 1). In an embodiment, the electronic device **200** may not include at least one (e.g., the key input device **217** or the light emitting element (not illustrated)) of the components or may further include any other component.

[0061] The display **120** may be visually exposed through a corresponding portion of the front plate **121**. For example, at least a portion of the display **201** may be visually exposed through the front plate **202** including the first surface **210A**, and the first area **210D** of the third surface **210C**. The display **201** may be disposed on a rear surface of the front plate **202**.

[0062] The corners of the display **201** may be formed to have substantially the same shape as an adjacent outer shape of the front plate **102**. In various embodiments, to increase an area, in which the display **201** is exposed, an interval between an outer side of the display **201** and an outer side of

the front plate **202** may be formed to be substantially the same.

[0063] A surface (or the front plate **202**) of the housing **110** may include a screen display area that is formed as the display **201** is visually exposed. For example, the screen display area may include the first surface **210A**, and the first areas **210D** on the side surface.

[0064] In various embodiments, the screen display areas **210A** and **210D** may include sensing areas (not illustrated) that is configured to acquire the user's biometric information. Here, the meaning of "the screen display areas **210A** and **210D** including sensing areas" may be understood as at least a portion of the sensing areas overlapping the screen display areas **210A** and **210D**. For example, the sensing area (not illustrated) may mean an area, in which visual information may be displayed by the display **201** as in the other areas of the screen display areas **210A** and **210D**, and the user's biometric information (e.g., fingerprint) may be additionally acquired.

[0065] The screen display areas **210A** and **210D** of the display **201** may include an area, in which the first camera module **205** (e.g., a punch hole camera) is visually exposed. For example, at least a portion of a periphery of an area, in which the first camera module **205** is visually exposed may be surrounded by the screen display areas **210A** and **210D**.

[0066] In various embodiments, the display **201** may be configured such that at least one of an audio module (not illustrated), a sensor module (not illustrated), a camera module (e.g., the first camera module **205**), and a light emitting element (not illustrated) is disposed on rear surfaces of the screen display areas **210A** and **210D**. For example, the electronic device **200** may be configured such that the first camera module **205** (e.g., an under display camera (UDC)) is disposed on a rear surface (e.g., a surface that faces the $-z$ -axis direction) of the first surface **210A** (e.g., the front surface) to face the first surface **210A**. For example, the first camera module **205** may be disposed under the display **201**, and may not be visually exposed to the screen display areas **210A** and **210D**.

[0067] In various embodiments, when the first camera module **205** is configured as an under display camera, an area of the display **201**, which faces the first camera module **205**, is a portion of a display area that displays contents, and may be formed as a transmissive area having a specified transmittance. For example, the transmissive area may be formed to have a transmittance in a range of about 5% to about 50%. The transmissive area may include an area that overlaps an effective area (e.g., a field of view (FOV) area) of the first camera module **205**, through which light for generating an image passes by forming an image by an image sensor. For example, the transmissive area of the display **201** may include an area having a lower pixel density and/or a lower wiring line density than those of the surrounding area.

[0068] The display **201** may be coupled, directly or indirectly, to a touch sensing circuit, a pressure sensor that may measure an intensity (or pressure) of a touch, and/or a digitizer that detects a magnetic stylus pen or may be disposed to be coupled or adjacent thereto.

[0069] The audio module **203**, **204**, and **207** may include microphone holes **203** and **204** and a speaker hole **207**.

[0070] The microphone holes **203** and **204** may include a first microphone hole **203** that is formed in a partial area of the third surface **210C** and a second microphone hole **204** that is formed in a partial area of the second surface **210B**. A microphone (not illustrated) for acquiring external sound may be disposed in the microphone holes **203** and **204**. The microphone may include a plurality of microphones to sense a direction of sound.

[0071] The second microphone hole **204** formed in a partial area of the second surface **210B** may be disposed adjacent to the camera modules **205**, **212**, and **213**. For example, the second microphone hole **204** may acquire sound when the camera modules **205**, **212**, and **213** are executed, or may acquire sound when other functions are executed.

[0072] The speaker hole **207** may include an external speaker hole **207** and a receiver hole (not illustrated) for a call. The external speaker hole **207** may be formed at a portion of the third surface **210C** of the electronic device **200**. In an embodiment, the external speaker hole **207** may be implemented as one hole with the microphone hole **203**. Although not illustrated, a call receiver

hole (not illustrated) may be formed at another portion of the third surface **210C**. For example, the call receiver hole may be formed at a portion (e.g., a portion that faces the $-y$ -axis direction) of the third surface **210C**, on which the external speaker hole **207** is formed, which portion faces another portion of the third surface **210C**. According to various embodiments, the call receiver hole may not be formed at a portion of the third surface **210C**, but may be formed by a space (e.g., the space “H” of FIG. 5) between the front plate **202** (or the display **201**) and the side bezel structure **218**.

[0073] The electronic device **200** may include at least one speaker (not illustrated) that is configured to output sound to the outside of the housing **210** through the external speaker hole **207** or the call receiver hole (not illustrated). According to various embodiments, the speaker may include a piezo speaker, from which the speaker hole **207** is omitted.

[0074] The sensor module (not illustrated) may generate an electrical signal or a data value that corresponds to an internal operation state of the electronic device **200** or an external environment state. For example, the sensor module **176** may include at least one or more of a proximity sensor, a HRM sensor, a fingerprint sensor, a gesture sensor, a gyro sensor, a barometric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illumination sensor.

[0075] The camera modules **205**, **212**, and **213** may include a first camera module **205** that is exposed through the first surface **210A** of the electronic device **200**, a second camera module **212** that is exposed through the second surface **210B**, and/or a flash **213**. Each of the first camera module **205** and the second camera module **212** may include one or more lenses, an image sensor, and/or an image signal processor. The flash **213** may include, for example, a light emitting diode or a xenon lamp.

[0076] The first camera module **205** may be visually exposed through a portion of the screen display areas **210A** and **210D** of the display **201**. For example, the first camera module **205** (e.g., a punch hole camera) may be visually exposed to a partial area of the screen display areas **210A** and **210D** through an opening (not illustrated) that is formed at a portion of the display **201**. In various embodiments, the first camera module **205** (e.g., an under-display camera) may be disposed on a rear surface of the display **201**, and may not be visually exposed to the screen display areas **210A** and **210D**.

[0077] The second camera module **212** may include a plurality of camera devices (e.g., a dual camera, a triple camera, or a quad camera). However, the second camera module **212** is not necessarily limited to including a plurality of camera devices, and may include only one camera device.

[0078] The key input device **217** may be disposed on the third surface **210C** of the housing **210**. In various embodiments, the electronic device **200** may not include some or all of the key input devices **217**, and the key input device **217** that is not included may be implemented in the form of a soft key on the display **201**.

[0079] The connector hole **208** may accommodate the connector. The connector hole **208** may be disposed on the third surface **210C** of the housing **210**. For example, the connector hole **208** may be disposed on the third surface **210C** to be adjacent to at least a portion of the audio module (e.g., the microphone hole **203** and the speaker hole **207**). In various embodiments, the electronic device **200** may include a first connector hole **208** that may accommodate a connector (e.g., a USB connector) for transmitting/receiving electric power and/or data to/from an external electronic device, and/or a second connector hole (e.g., a second connector hole (not illustrated) that may accommodate the connector (e.g., an earphone jack)) for transmitting/receiving an audio signal to/from an external electronic device.

[0080] The electronic device **200** may include a light emitting element (not illustrated). For example, the light emitting element (not illustrated) may be disposed on the first surface **210A** of the housing **210**. The light emitting element (not illustrated) may provide state information of the electronic device **200** in an optical form. In various embodiments, the light emitting element (not

illustrated) may provide a light source that is operated in conjunction with the operation of the first camera module **205**. For example, the light emitting element (not illustrated) may include an LED, an IR LED, and/or a xenon lamp.

[0081] FIG. **3** is an exploded perspective view of an electronic device according to an embodiment.

[0082] FIG. **3** is a view illustrating a state, in which the electronic device is disassembled with reference to the rear perspective view of FIG. **2B**.

[0083] Referring to FIG. **3**, an electronic device **200** according to an embodiment may include a front plate **220** (e.g., the front plate **202** of FIG. **2A**), a display **230** (e.g., the display **201** of FIG. **2A**), a side member **240** (e.g., the side bezel structure **218** of FIG. **2A**), a printed circuit board **250**, a rear case **260**, a battery **270**, a rear plate **280** (e.g., the rear plate **211** of FIG. **2B**), a camera module **291** and **292**, a speaker module **293**, and an antenna (not illustrated). In various embodiments, the electronic device **200** may omit at least some of the above components or may further include another component.

[0084] At least some of the components of the electronic device **200** illustrated in FIG. **3** may be the same as or similar to the components of the electronic device **200** illustrated in FIGS. **2A** and **2B**, and hereinafter, repeated descriptions thereof will be omitted.

[0085] The front plate **220**, the side member **240**, and the rear plate **280** may constitute a housing (e.g., the housing **210** of FIGS. **2A** and **2B**) of the electronic device **200**. For example, the housing **210** of the electronic device **200** may refer to a structure, in which a specific accommodation space is formed or defined in an interior thereof by combining the front plate **220**, the side member **240**, and the rear plate **280**.

[0086] The front plate **220** and the display **230** may be coupled, directly or indirectly, to the side member **240**. For example, the front plate **220** and the display **230** may be disposed under the side member **240** with reference to FIG. **3**. The front plate **220** and the display **230** may be located in the +z-axis direction from the side member **240**. For example, the display **230** may be coupled under the side member **240**, and the front plate **220** may be coupled under the display **230**. The front plate **220** may define a portion of an outer surface (or an exterior) of the electronic device **200**. The display **230** may be disposed between the front plate **220** and the side member **240** to be located in an interior of the electronic device **200**.

[0087] The side member **240** may be disposed between the display **230** and the rear plate **280**. For example, the side member **240** may be configured to surround the space between the rear plate **280** and the display **230**.

[0088] The side member **240** may include a frame part **241** that defines a portion of a side surface (e.g., the third surface **210C** of FIG. **2A**) of the electronic device **200**, and a plate part **242** that extends inward from the frame part **241**.

[0089] The plate part **242** may be disposed in an interior of the frame part **241** to be surrounded by the frame part **241**. The plate part **242** may be connected to the frame part **241**, or may be integrally formed with the frame part **241**. The plate part **242** may be formed of a metal material and/or a non-metal material (e.g., polymer). In an embodiment, the plate part **242** may support other components included in the electronic device **200**. For example, at least one of the display **230**, the printed circuit board **250**, the rear case **260**, and the battery **270** may be disposed in the plate part **242**. The display **230** may be coupled, directly or indirectly, to one surface (e.g., a surface that faces the +z-axis direction) of the plate part **242**, and the printed circuit board **250** may be coupled, directly or indirectly, to a surface (e.g., a surface that faces the -z-axis direction) that faces an opposite direction to the one surface.

[0090] A processor (e.g., the processor **120** of FIG. **1**), a memory (e.g., the memory **130** of FIG. **1**), and/or an interface (e.g., the interface **177** of FIG. **1**) may be mounted, directly or indirectly, on the printed circuit board **250**. For example, the processor may include one or more of a central processing unit, an application processor, a graphic processing device, an image signal processor, a sensor hub processor, or a communication processor. The memory may include, for example, a

volatile memory or a nonvolatile memory. The interface may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, and/or an audio interface. The interface may electrically or physically connect the electronic device **200** to an external electronic device and may include a USB connector, an SD card/MMC connector, or an audio connector.

[0091] The rear case **260** may be disposed between the rear plate **280** and the plate part **242**. The rear case **260** may be coupled, directly or indirectly, to the side member **240** to overlap at least a portion of the printed circuit board **250**. For example, the rear case **260** may face the plate part **242** with the printed circuit board **250** interposed therebetween.

[0092] The battery **270** (e.g., the battery **189** of FIG. **1**) may supply power to at least one component of the electronic device **200**. For example, a battery **270** may include a non-rechargeable primary battery, a rechargeable secondary battery, or a fuel cell. At least a portion of the battery **270** may be disposed on substantially the same plane as the printed circuit board **250**. The battery **270** may be integrally disposed in an interior of the electronic device **200**, or may be disposed in the electronic device **200** to be detachable.

[0093] The rear plate **280** may be coupled, directly or indirectly, to the side member **240**. The rear plate **280** may include a rear camera area **281** for supporting the function/operation of the rear camera (e.g., the second camera module **292**). The rear camera area **281** may be formed on a surface (e.g., the rear surface **210B** of FIG. **2B**) of the rear plate **280**. The camera area **281** may be formed such that at least a portion thereof is transparent so that external light is input to the lens of the second camera module **292**. At least a portion of the camera area **281** may protrude from the surface of the rear plate **280** by a specific height. However, the disclosure is not limited thereto, and the camera area **281** may define substantially the same plane as the surface of the rear plate **280**.

[0094] The first camera module **291** (e.g., the first camera module **205** of FIG. **2A**) may be disposed in the side member **240** (e.g., the plate part **242**) so that the lens may receive external light through a partial area of the front plate **220** (e.g., the front surface **210A** of FIG. **2A**). For example, the lens of the first camera module **291** may be visually exposed through a partial area of the front plate **220**. A front camera area **231** (e.g., an opening area or a light transmission area) corresponding to the first camera module **291** may be formed in the display **230**.

[0095] The second camera module **292** (e.g., the second camera module **212** of FIG. **2B**) may be disposed in the printed circuit board **250** so that the lens may receive external light through the rear camera area **281** of the rear plate **280** (e.g., the rear surface **210B** of FIG. **2B**) of the electronic device **200**. For example, the lens of the second camera module **212** may be visually exposed to the camera area **281**. The second camera module **292** may be disposed at at least a portion of an interior space that is formed in the housing (e.g., the housing **210** of FIGS. **2A** and **2B**) of the electronic device **200**, and may be electrically connected to the printed circuit board **250** through a connecting member (e.g., a connector).

[0096] The speaker module **293** (e.g., the sound output module **155** of FIG. **1**) may be electrically connected to the printed circuit board **250**. The speaker module **293** may be supported by, directly or indirectly, a plate part **242** of the side member **240**, and may be surrounded by the printed circuit board **250**.

[0097] An antenna (not illustrated) (e.g., the antenna module **197** of FIG. **1**) may be disposed between the rear plate **280** and the battery **270**. The antenna (not illustrated) may include, for example, a near field communication (NFC) antenna, an antenna for wireless charging, and/or a magnetic secure transmission (MST) antenna. For example, the antenna (not illustrated) may perform short range communication with an external device or may wirelessly transmit/receive power necessary for charging.

[0098] FIG. **4A** is a rear perspective view of an electronic device according to an embodiment. FIG. **4B** is a rear perspective view of an electronic device according to an embodiment.

[0099] FIG. **4B** illustrates a state, in which the rear case is separated from the electronic device

illustrated in FIG. 4B.

[0100] Referring to FIGS. 4A and 4B, an electronic device 300 (e.g., the electronic device 200 of FIG. 3) according to an embodiment may include a side member 310 (e.g., the side member 240 of FIG. 3), a circuit board 320 (e.g., the printed circuit board 250 of FIG. 3), a rear case 330 (e.g., the rear case 260 of FIG. 3), a connecting member 340, an elastic structure 350, and a battery 391 (e.g., the battery 270 of FIG. 3).

[0101] At least some of the components of the electronic device 300 illustrated in FIGS. 4A and 4B may be substantially the same as or similar to the components of the electronic device 300 illustrated in FIG. 3, and hereinafter, repeated descriptions thereof will be omitted.

[0102] The side member 310 may constitute the housing 301 (e.g., the housing 210 of FIGS. 2A and 2B) of the electronic device 300, or may be a portion of the housing 301. For example, the housing 301 of the electronic device 300 may be provided in a structure, in which the side member 310, the front plate (e.g., the front plate 220 of FIG. 3) and the rear plate (e.g., the rear plate 280 of FIG. 3) are coupled to each other. For example, FIG. 4A may be a view, in which the front plate and the rear plate of the electronic device 300 are omitted and only the side member 310 is illustrated.

[0103] The side member 310 may include a plate part 312 (e.g., a plate part 242 of FIG. 3), in which the circuit board 320, the battery 391, and the elastic structure 350 are disposed, and a frame part 311 (e.g., a frame part 241 of FIG. 3) that surrounds a periphery of the plate part 312. The rear case 330 may be coupled to the plate part 312. For example, the circuit board 320, the battery 391, and the elastic structure 350 may be disposed on one side (e.g., the -z-axis direction) of the plate part 312, and the rear case 330 may be coupled to overlap the circuit board 320. The plate part 312 may include a sidewall 314 that forms or defines a space, in which the battery 391 is accommodated. For example, the sidewall 314 may extend to surround a periphery of the battery 391. A portion (e.g., a portion that faces the -y-axis direction) of the sidewall 314 may face the elastic structure 350. According to various embodiments, the plate part 312 may be referred to as a front case. FIG. 4B may be a view, in which a frame part 311 of the side member 310 is omitted.

[0104] The circuit board 320 may be disposed in the plate part 312 of the side member 310. For example, the circuit board 320 may be coupled to the plate part 312 through various fastening means (e.g., screws). For example, the circuit board 320 may be a printed circuit board (PCB). The circuit board 320 may include a first circuit board 321 that is located at an upper end (e.g., the +y-axis direction) of the electronic device 300, and a second circuit board 322 that is located at a lower end (e.g., the -y-axis direction) of the electronic device 300. For example, the first circuit board 321 may be located in the +y-axis direction with respect to the battery 391, and the second circuit board 322 may be located in the -y-axis direction with respect to the battery 391. The first circuit board 321 may be referenced as a printed circuit board (e.g., the printed circuit board 250 of FIG. 3) of the electronic device 300 illustrated in FIG. 3. The first circuit board 321 and the second circuit board 322 may be electrically connected to each other through a connecting member 340. According to various embodiments, the first circuit board 321 may be referred to as an upper PCB or a main PCB, and the second circuit board 322 may be referred to as a lower PCB or a sub-PCB.

[0105] The rear case 330 may be coupled to the plate part 312 of the side member 310. For example, the rear case 330 may be coupled to the plate part 312 to cover the circuit board 320. The rear case 330 may overlap the circuit board 320 with respect to the rear ward direction (e.g., the -z-axis direction) of the electronic device 300. The rear case 330 may include a first rear case 331 that is located at an upper end (e.g., the +y-axis direction) of the electronic device 300, and a second rear case 332 that is located at a lower end (e.g., the -y-axis direction) of the electronic device 300. For example, the first rear case 331 may be located in the +y-axis direction with respect to the battery 391, and the second rear case 332 may be located in the -y-axis direction with respect to the battery 391. The first rear case 331 may be referenced as a rear case (e.g., the rear case 260 of FIG. 3) of the electronic device 300 illustrated in FIG. 3. In various embodiments, the first rear

case **331** and the second rear case **332** may be connected to each other, and may be integrally formed.

[0106] The first rear case **331** may face at least a portion of the first circuit board **321**, and the second rear case **332** may face at least a portion of the second circuit board **322**. For example, the first rear case **331** may be coupled to the plate part **312** to cover the first circuit board **321**, and the second rear case **332** may be coupled to the plate part **312** to cover the second circuit board **322**. The first rear case **331** may overlap the first circuit board **321** in a rearward direction of the electronic device **300**. For example, the first circuit board **321** may be disposed in a space between a partial area of the plate part **312** and the first rear case. The second rear case **332** may overlap the second circuit board **322** and the elastic structure **350** in the rearward direction of the electronic device **300**. For example, the elastic structure **350** and the second circuit board **322** may be disposed in a space between a partial area of the plate part **312** and the second rear case **332**. When the rear case **330** is viewed from above with reference to FIG. 4A, the first circuit board **321** and one end of the connecting member **340** may be located under the first rear case **331**, and the elastic structure **350**, the second circuit board **322**, and an opposite end of the connecting member **340** may be located under the second rear case **332**.

[0107] The connecting member **340** may electrically connect the first circuit board **321** and the second circuit board **322**. One end (e.g., an end in the +y-axis direction) of the connecting member **340** may be connected to the first circuit board **321**, and an opposite end (e.g., an end in the -y-axis direction) of the connecting member **340** may be connected to the second circuit board **322**. For example, opposite ends of the connecting member **340** may be formed as connectors **341**. The connecting member **340** may extend in the lengthwise direction (e.g., the y-axis direction) of the electronic device **300**, and may cross the battery **391**. The connecting member **340** may extend from the first circuit board **321** located on one side (e.g., the +y-axis direction) of the battery **391** to the second circuit board **322** located on an opposite side (e.g., the -y-axis direction) of the battery **391**.

[0108] The connecting member **340** may be formed to have a margin of a specific length to prevent or reduce separation from the first circuit board **321** and the second circuit board **322**. The connecting member **340** may be connected to the first circuit board **321** and the second circuit board **322** while at least a portion thereof is bent. The connecting member **340** may be formed of a conductive material having flexibility, and may be moved in a specific range in the lengthwise direction of the connecting member **340** by shaking the electronic device **300** or by an external force applied to the electronic device **300**. For example, the connecting member **340** may include a flexible printed circuit board (FPCB). At least a portion of the connecting member **340** may be supported, directly or indirectly, by the elastic structure **350** coupled to the plate part **312**, and the movement of the connecting member **340** may be guided by the elastic driving of the elastic structure **350**.

[0109] The elastic structure **350** may be coupled, directly or indirectly, to the plate part **312**. The elastic structure **350** may support a partial section of the connecting member **340**. The elastic structure **350** may guide the movement and/or deformation of the connecting member **340** by elastically driving some components with an external force, and may formalize the bent shape of the connecting member **340**. Hereinafter, the structure and driving of the elastic structure **350** will be described in detail with reference to FIGS. 5, 6, and 7.

[0110] FIG. 5 is a view illustrating an elastic structure of an electronic device according to an embodiment. FIG. 6 is a view illustrating a connecting member and an elastic structure of an electronic device according to an embodiment. FIG. 7 is a view illustrating driving of an elastic structure of an electronic device according to an embodiment.

[0111] FIG. 5 illustrates a front view, a left side view, and a right side view of the elastic structure. For example, FIG. 5 illustrates a basic state of the elastic structure, to which an external force is not illustrated. FIG. 6 is a view, in which other components than the connecting member and the elastic

structure in FIG. 4B are omitted. FIG. 7 is a view illustrating an operation of deforming the elastic structure in correspondence to an external force applied to the elastic structure. For example, FIG. 7 illustrates views that illustrate an operation of deforming the elastic structure from a basic state by an external force in FIG. 7, when viewed from a left side and a right side.

[0112] Referring to FIGS. 5 and 6, the elastic structure 350 of the electronic device 300 according to an embodiment may include a holder 360, and a rod 370 that is coupled, directly or indirectly, to the holder 360 to be rotatable.

[0113] The holder 360 may be coupled, directly or indirectly, to a side member (e.g., the side member 310 of FIGS. 4A and 4B) of the electronic device 300. The holder 360 may include a first base part 364, a first coupling part 365 that extends from one end of the first base part 364, and a second coupling part 366 that extends from an opposite end of the first base part 364. The first base part 364 may be coupled to a plate part (e.g., the plate part 312 of FIG. 4B). Opposite ends of the rod 370 may be coupled to the first coupling part 365 and the second coupling part 366 to be rotatable, respectively. For example, the first extension part 375 of the rod 370 may be coupled to the first coupling part 365 to be rotatable, and the second extension part 376 of the rod 370 may be coupled to the second coupling part 366 to be rotatable.

[0114] The first base part 364 of the holder 360 may include a first portion 361 that is seated on, attached to, or coupled to the plate part 312, a second portion 362 that connects the first portion 361 and the first coupling part 365, and a third portion 363 that connects the first portion 361 and the second coupling part 366. The first portion 361 may be parallel to one surface of the plate part 312, and the second portion 362 and the third portion 363 may extend from opposite ends of the first portion 361. For example, the first portion 361 of the first base part 364 may be parallel to the second circuit board (e.g., the second circuit board 322 of FIG. 9A) or the rear plate (e.g., the rear plate 380 of FIG. 9A) (e.g., see FIG. 9A). For example, the first coupling part 365 may extend from the second portion 362 of the first base part 364, and the second coupling part 366 may extend from the third portion 363 of the first base part 364.

[0115] The coupling parts 365 and 366 of the holder 360 may have elasticity to apply a specific pre-pressure to the extension parts 375 and 376 of the rod 370. The first coupling part 365 and the second coupling part 366 may be bent in correspondence to the shapes of the first extension part 375 and the second extension part 376, and the first extension part 375 and the second extension part 376 may be elastically fitted into the interiors of the first coupling part 365 and the second coupling part 366, respectively. For example, recesses of shapes corresponding to the first extension part 375 and the second extension part 376 may be formed in the first coupling part 365 and the second coupling part 366, and the first extension part 375 and the second extension part 376 may be partially inserted into the corresponding recesses of the first coupling part 365 and the second coupling part 366, respectively.

[0116] The holder 360 may be formed in a shape, in which the first coupling part 365 and the second coupling part 366 are located at different heights from the first base part 364. The first coupling part 365 and the second coupling part 366 may be formed at positions that are spaced apart from the first portion 361 of the first base part 364 by the same distance in the first direction {circle around (1)}. The first direction {circle around (1)} may be a direction (e.g., the +y-axis direction) that is parallel to the first portion 361, and in which the coupling parts 365 and 366 extend from the first base part 364. For example, the distance between the first portion 361 and the first coupling part 365 with respect to the first direction {circle around (1)} may be the same as the distance between the first portion 361 and the second coupling part 366. For example, the first coupling part 365 and the second coupling part 366 may be spaced apart from a first periphery of the first base part 364 (e.g., the first portion 361) by the same distance in the first direction {circle around (1)}. The first periphery is a periphery of the first port 361, which faces the second direction {circle around (2)} (e.g., the -y-axis direction) that is opposite to the first direction {circle around (1)}. The first coupling part 365 and the second coupling part 366 may be formed at positions that

are spaced apart from the first portion **361** of the first base part **364** by different distances in a third direction {circle around (3)} that is perpendicular to the first direction {circle around (1)}. For example, the third direction {circle around (3)} may be a direction (e.g., the $-z$ -axis direction) that is substantially perpendicular to one surface of the first port **361**, and the third direction {circle around (3)} may be perpendicular to the first direction {circle around (1)} and the second direction {circle around (2)}. The first coupling part **365** may be spaced apart from the first portion **361** by a second distance in the third direction {circle around (3)}, and the second coupling part **366** may be spaced apart from the first portion **361** by a second distance that is greater than the first distance in the third direction {circle around (3)}. As the first extension part **375** and the second extension part **376** are coupled to be first coupling part **365** and the second coupling part **366** to be rotatable, respectively, a first rotation axis **R1** and a second rotation axis **R2** of opposite ends of the rod **370** may be arranged in parallel with respect to the third direction {circle around (3)}, and a height difference (e.g., a difference between a first height **H1** and a second height **H2** of FIG. 7) may occur along the third direction {circle around (3)} between the first rotation axis **R1** and the second rotation axis **R2**.

[0117] Opposite ends of the rod **370** may be coupled to the holder **360** to be rotatable, respectively, and may have a specific elasticity. The rod **370** may include a second base part **374**, a first extension part **375** that extends from one end of the second base part **374**, and a second extension part **376** that extends from an opposite end of the second base part **374**. The rod **370** may be formed or provided in a shape, in which the first extension part **375** and the second extension part **376** may be coupled to the first coupling part **365** and the second coupling part **366** of the holder **360**, respectively, in correspondence to the shape of the holder **360**. The first extension part **375** may define or form the first rotation axis **R1** as it is coupled to the first coupling part **365** of the holder **360** to be rotatable. The second extension part **376** may define or form the second rotation axis **R2** as it is coupled to the second coupling part **366** of the holder **360** to be rotatable. For example, the first rotation axis **R1** of the rod **370** may be an imaginary straight line that passes through the center of the first extension part **375**, and the first rotation axis **R1** of the rod **370** may be an imaginary straight line that passes through the center of the second extension part **376**.

[0118] The second base part **374** of the rod **370** may include a fourth portion **371** that extends long in the direction of the rotation axes **R1** and **R2**, a fifth portion **372** that connects the fourth portion **371** to the first extension part **375**, and a sixth portion **373** that connects the fourth portion **371** to the second extension part **376**. For example, the first extension part **375** may extend from the fifth portion **372** substantially perpendicularly thereto. The second extension part **376** may extend from the sixth portion **373** substantially perpendicularly thereto. The fourth portion **371** of the second base part **374** may extend parallel to a width “W” direction (e.g., the x -axis direction) of the connecting member **340**. Each of the fifth and sixth portions **372** and **373** may be substantially perpendicular to the fourth portion **371**, but may extend in different lengths and different directions from opposite ends of the fourth portion **371**. The fifth portion **372** and the sixth portion **373** may extend in correspondence to the height difference between the first coupling part **365** and the second coupling part **366**, into which the first extension part **375** and the second extension part **376** are inserted.

[0119] The rod **370** may be formed in a shape, in which the first extension part **375** and the second extension part **376** are not aligned with and misaligned from each other in a direction (directions of the rotation axes **R1** and **R2** or the x -axis direction), in which the fourth portion **371** of the second base part **374** extends. As illustrated in FIG. 5, in a basic state, in which no external force is applied to the elastic structure **350**, the first extension part **375** and the second extension part **376** may be formed at positions that are spaced apart from the fourth portion **371** of the second base part **374** by the same distance in the first direction {circle around (1)}. For example, the distance between the fourth portion **371** and the first extension part **375** with respect to the first direction {circle around (1)} may be the same as the distance between the fourth portion **371** and the second extension part

376. For example, the first extension part **375** and the second extension part **376** may be spaced apart from the fourth portion **371** by the same distance in the first direction {circle around (1)}. As illustrated in FIG. 5, in a basic state, in which no external force is applied to the elastic structure **350**, the first extension part **375** and the second extension part **376** may be formed at positions that are spaced apart from the fourth portion **371** of the second base part **374** by different distances in an opposite direction to the third direction {circle around (3)}. The first extension part **375** may be spaced apart from the fourth portion **371** by a third distance in an opposite direction to the third direction {circle around (3)}, and the second extension part **376** may be spaced apart from the first portion **361** in an opposite direction to the third direction {circle around (3)} by a fourth distance that is smaller than the third distance. For example, the distances between the fourth portion **371**, the first extension part **375**, and the second extension part **376** may be the distances between the central axes thereof. As the first extension part **375** and the second extension part **376** of the rod **370** are coupled to the first coupling part **365** and the second coupling part **366** of the holder **360** to be rotatable, respectively, the first rotation axis **R1** and the second rotation axis **R2** may be located on an imaginary line “L” that is parallel to the third direction {circle around (3)}, and the second rotation axis **R2** may be spaced apart from the first rotation axis **R1** by a specific distance (e.g., a difference between the first height **H1** and the second height **H2**) in the third direction {circle around (3)}.

[0120] The connecting member **340** may be connected to the elastic structure **350**. The connecting member **340** may extend across a space between the rod **370** and the holder **360**. For example, an opening **351** may be formed between the holder **360** and the rod **370** of the elastic structure **350**. The connecting member **340** may pass through the opening **351** of the elastic structure **350**. For example, at least a portion of the connecting member **340** may contact the rod **370** (e.g., the fourth portion **371** of the rod **370**), and may extend through the opening **351**. With reference to FIG. 6, the connecting member **340** may be connected to or assembled in the elastic structure **350** such that the rod **370** is located at a lower portion (e.g., the third direction {circle around (3)}) of a portion of the connecting member **340**, and the holder is located at an upper portion (e.g., the third direction {circle around (3)}) of a portion of the connecting member **340**.

[0121] Referring to FIGS. 5, 6, and 7, the elastic structure **350** may control and optimize the movement of the connecting member **340** by using the torsional elasticity of the rod **370**. For example, the elastic structure **350** may be a spring structure that uses a repulsive force generated when the rod **370** is twisted by an external force. When an external force is applied to the elastic structure **350** in a basic state, opposite ends of the rod **370** may be rotated with respect to the holder **360** around the first rotation axis **R1** and the second rotation axis **R2**, respectively, and a torsional displacement (or a torsional deformation) may occur as the rod **370** is twisted due to the height difference between the first rotation axis **R1** and the second rotation axis **R2**. The rod **370** in a twisted state may have a restoring force to return to its original state, and the elastic structure **350** may return to a basic state by the restoring force of the rod **370** when the applied external force is removed. In a state, in which no external force is applied to the electronic device **300**, the connecting member **340** may be pulled in the second direction {circle around (2)} by the elastic structure **350** having a restoring force for maintaining a basic state, and may be bent in a specific shape in an interior of the electronic device **300**. For example, when an external force that is greater than the restoring force of the elastic structure **350** is not applied to the electronic device **300**, the connecting member **340** and the elastic structure **350** may be maintain in a state illustrated in FIG. 6 or a state indicated by a dotted line in FIG. 7.

[0122] The rod **370** may be twisted as opposite ends thereof are rotated around the first rotation axis **R1** and the second rotation axis **R2** located at different heights, respectively. For example, the first rotation axis **R1** and the second rotation axis **R2** may be aligned on the same line “L” with respect to the third direction {circle around (3)}, but may be spaced apart from each other. The first rotation axis **R1** may be located at a first height **H1** in the third direction {circle around (3)} from

the first portion **361** of the first base part **364**. The second rotation axis **R2** may be located at a second height **H2** that is greater than the first height **H1** in the third direction {circle around (3)} from the first portion **361** of the first base part **364**. The first rotation axis **R1** and the second rotation axis **R2** may be spaced apart from each other by the difference between the first height **H1** and the second height **H2** on the imaging line “L” that is parallel to the third direction {circle around (3)}.

[0123] Referring to FIGS. **6** and **7**, as the connecting member **340** is moved in the first direction {circle around (1)} or the second direction {circle around (2)}, an external force may be applied to the elastic structure **350** or the applied external force may be removed. When the connecting member **340** is moved in the first direction {circle around (1)} as an external force is applied to the electronic device **300**, an external force may be applied to the rod **370** of the elastic structure **350** in a basic state in the first direction {circle around (1)} by the connecting member **340**, and the rod **370** may be twisted while opposite ends **375** and **376** thereof are rotated in a first rotational direction (e.g., a direction that becomes more distant from the first base part **364** of the holder **360**) around the first rotation axis **R1** and the second rotation axis **R2**. When the connecting member **340** is moved in the second direction {circle around (2)} as the external force applied to the electronic device **300** is removed, the elastic structure **350** may return to the basic state while opposite ends **375** and **376** of the rod **370** in a twisted state are rotated with respect to the holder **360** in a second rotational direction (e.g., a direction that becomes closer to the first base part **364** of the holder **360**) that is opposite to the first rotational direction around the first rotation axis **R1** and the second rotation axis **R2**. For example, in the basic state of the elastic structure **350**, that is, when an external force is not applied to the elastic structure **350**, the elastic structure **350** may have a pattern or a shape that is maintained basically, and when the external force applied to the elastic structure **350** is smaller than a magnitude of a force, by which a torsional displacement may occur in the rod **370** (e.g., the external force is smaller than a torsional repulsive force of the rod **370**), the elastic structure **350** may be maintained in the basic state.

[0124] When the rod **370** is rotated in the first rotational direction as an external force is applied to the elastic structure **350** in the basic state in the first direction {circle around (1)}, an angle, by which the fifth portion **372** connected to the first extension part **375** is rotated, and an angle, by which the sixth portion **373** connected to the second extension part **376** is rotated, may be different due to the height difference between the first rotation axis **R1** and the second rotation axis **R2**. For example, in an operation of moving the connecting member **340** in the first direction {circle around (1)} by a specific distance, the fifth portion **372** connected to the first extension part **375** rotated around the first rotation axis **R1** may be rotated in the first rotational direction by a first angle **A1**, and the sixth portion **373** connected to the second extension part **376** rotated around the second rotation axis **R2** may be rotated in the first rotational direction by a second angle **A2** that is greater than the first angle **A1**. As the fifth portion **372** and the sixth portion **373** of the second base part **374** are rotated at different angles, a torsional displacement may occur in the rod **370**.

[0125] The structure of the elastic structure **350** illustrated in FIGS. **5**, **6**, and **7** is illustrated by way of example, and the structure of the elastic structure **350** according to embodiments disclosed in the disclosure is not limited to those of FIGS. **5**, **6**, and **7**, and may be implemented in various forms. Various embodiments of the elastic structure **350** are described below with reference to FIGS. **13** to **18**.

[0126] FIG. **8** is a view illustrating a rear case and a connecting member of an electronic device according to an embodiment. FIG. **9A** is a cross-sectional view of an electronic device according to an embodiment. FIG. **9B** is a cross-sectional view of an electronic device according to an embodiment.

[0127] FIG. **8** illustrates a plan view of a portion of an electronic device, in which the connecting member and the rear case are disposed, and a view illustrating a shape of the rear case. FIG. **9A** is a view illustrating cross section A-A' of the electronic device according to an embodiment illustrated

in FIG. 8. FIG. 9B is a view illustrating cross section B-B' and cross-section C-C' of the electronic device according to an embodiment illustrated in FIG. 8.

[0128] Referring to FIGS. 8, 9A, and 9B, an electronic device 300 according to an embodiment may include a side member 310, a second circuit board 322, a second rear case 332, a connecting member 340, an elastic structure 350, a rear plate 380 (e.g., the rear plate 380 of FIG. 3), and a battery 391.

[0129] At least some of the components of the electronic device 300 illustrated in FIGS. 8, 9A, and 9B may be substantially the same as or similar to the components of the electronic device 300 illustrated in FIGS. 4A and 4B and/or the elastic structure 350 illustrated in FIGS. 5 to 7, and repeated descriptions thereof will be omitted. For example, FIGS. 9A and 9B may be views illustrating a cross section of the electronic device 300 when the elastic structure 350 is in a basic state (e.g., the state of the elastic structure 350 illustrated in FIGS. 5 and 6).

[0130] The side member 310 may include a frame part 311 and a plate part 312. A second circuit board 322, a second rear case 332, and an elastic structure 350 may be coupled to the plate part 312. A through-hole 315 (e.g., a through-hole 315 of FIG. 13), through which the connecting member 340 passes, may be formed in the plate part 312. For example, the through-hole 315 may be formed on one side of a portion, at which the second circuit board 322 and the elastic structure 350 are coupled to each other. The connecting member 340 may extend from a space between the plate part 312 and the rear plate 380 to a space between the plate part 312 and the display (e.g., the display 230 of FIG. 3) through an opening (e.g., the opening 351 of FIG. 6) of the elastic structure 350 and the through-hole 315. The side member 310 may include a sidewall 314 (e.g., the sidewall 314 of FIG. 4B) that defines a space, in which the battery 391 is accommodated, and a portion (e.g., a portion that faces the -y-axis direction) of the sidewall 314 may face the elastic structure 350. A portion of the sidewall 314, which faces the elastic structure 350, may control or limit a driving range (e.g., a rotation range of the rod 370) of the elastic structure 350 while contacting the rod 370.

[0131] The second circuit board 322 and the elastic structure 350 may be coupled, directly or indirectly, to the plate part 312. The second circuit board 322 and the elastic structure 350 may be disposed between the plate part 312 and the second rear case 332. The second circuit board 322 and the holder 360 of the elastic structure 350 may be fixed to the plate part 312 through the first fixing member 392. For example, the holder 360 may be disposed between the plate part 312 and the second circuit board 322, and the first fixing member 392 may sequentially pass through the second circuit board 322 and the holder 360 to be fastened to the plate part 312. The first fixing member 392 may be a screw, but the disclosure is not limited thereto.

[0132] The second rear case 332 may be coupled to the plate part 312 to cover the second circuit board 322 and the elastic structure 350. The second rear case 332 may include a first protruding part 334 and a second protruding part 335 for guiding (or controlling) the movement and/or the shape of the connecting member 340.

[0133] The first protruding part 334 of the second rear case 332 may protrude in the first direction {circle around (1)} from the body part 333 of the rear case 330 to be located between the elastic structure 350 and the rear plate 380. The first protruding part 334 may be located in the third direction {circle around (3)} with respect to the elastic structure 350, and may cover a partial area of the connecting member 340 connected to the elastic structure 350. For example, when the second rear case 332 is viewed from a top with reference to FIG. 8, a portion of the connecting member 340 may be disposed under the first protruding part 334. The first protruding part 334 may be spaced apart from the elastic structure 350 so as not to collide with the rod 370 when the rod 370 of the elastic structure 350 is rotated. For example, a space, in which the connecting member 340 may be moved, may be formed or defined between the first protruding part 334 and the rod 370. The first protruding part 334 may be formed as a surface 334s, in which a surface that faces the connecting member 340 is inclined at a specific inclination. The first protruding part 334 may

prevent or limit the connecting member **340** from colliding with the rear plate **380** or pushing the rear plate **380** in the third direction {circle around (3)} when the connecting member **340** is moved in the first direction {circle around (1)} or the second direction {circle around (2)} by the external force applied to the electronic device **300**. For example, the connecting member **340** may be moved along the inclined surface **334s** of the first protruding part **334**.

[0134] The second protruding part **335** of the second rear case **332** may protrude in the first direction {circle around (1)} from the body part **333** of the rear case **330** to be partially located between the rod **370** of the elastic structure **350** and the plate part **312**. For example, at least a portion of the second protruding part **335** may be disposed within the opening **351** of the elastic structure **350**. The second protruding part **335** may be located to face the plate part **312**, and a portion of the connecting member **340** may be disposed in a space between the second protruding part **335** and the plate part **312**. The second protruding part **335** may formalize a shape, in which the connecting member **340** is bent. For example, the connecting member **340** may be bent into a specific shape while extending along a spacing space between the second protruding part **335** and the plate part **312**. The second protruding part **335** may guide or control the movement of the connecting member **340** so that the connecting member **340** is not rapidly dragged when the connecting member **340** is moved in the first direction {circle around (1)}. For example, as the connecting member **340** is moved between the second protruding part **335** and the plate part **312** while being bent in correspondence to the shape of the second protruding part **335**, a rapid movement thereof may be prevented or reduced.

[0135] According to the embodiment illustrated in FIG. **8**, the second rear case **332** may have a first cutaway portion **336** formed on opposite sides of the first protruding part **334**, and a second cutaway portion **337** may be formed between the pair of second protruding parts **335**. For example, the second protruding part **335** may overlap the first cutaway portion **336** on opposite sides of the first protruding part **334** in the third direction {circle around (3)}, and the first protruding part **334** may overlap the second cutaway portion **337** between the second protruding parts **335** in the third direction {circle around (3)}. The embodiment illustrated in FIG. **8** relates to the second rear case **332** manufactured through an injection process using a mold, and the cutaway portions **336** and **337** may be spaces for removing the mold in the process of separating or extracting the injection-molded product from the mold. When the second rear case **332** is manufactured through the injection process using the mold, production costs may be reduced. The manufacturing method and the shape of the second rear case **332** are not limited to the embodiment illustrated in FIG. **8**, and according to various embodiments, the first protruding part **334** and the second protruding part **335** may extend without the cutaway portions **336** and **337** (e.g., see FIGS. **10** and **11**).

[0136] The connecting member **340** may be bent into a specific shape while a portion thereof is supported by, directly or indirectly, the elastic structure **350**. A first end (e.g., an end in the +y-axis direction and an end in the first direction {circle around (1)} of the connecting member **340** may be connected to the first circuit board (e.g., the first circuit board **321** of FIG. **4A**) at an upper end of the electronic device **300**. A central portion (e.g., the remaining portion except opposite ends thereof) of the connecting member **340** may extend from the first end in the second direction {circle around (2)} across the battery **391** such that at least a portion thereof is supported by, directly or indirectly, the elastic structure **350**. A second end (e.g., an end in the -y-axis direction or an end in the second direction {circle around (2)} of the connecting member **340**, which faces an opposite side to the first end may be connected to the second circuit board **322** at a lower end of the electronic device **300**.

[0137] A central portion of the connecting member **340** may extend through the opening **351** of the elastic structure **350** and the through-hole **315** of the plate part **312** such that a portion thereof is located above the plate part **312** (e.g., in the third direction {circle around (3)} with respect to the plate part **312**) and another portion thereof is located under the plate part **312** (e.g., an opposite direction to the third direction {circle around (3)} with respect to the plate part **312**). For example,

at least a partial section of the central portion of the connecting member **340** may extend sequentially across a space between the battery **391** and the rear plate **380**, a space between the first protruding part **334** of the second rear case **332** and the rod **370**, the opening **351** of the elastic structure **350**, a space between the side wall **314** of the plate part **312** and the second protruding part **335** of the second rear case **332**, and the through-hole **315** of the plate part **312**.

[0138] The connecting member **340** may be maintained in a state, in which it is bent in a specific shape, in an interior of the electronic device **300** by the elastic structure **350** in a basic state. For example, when no external force is applied to the electronic device **300**, the elastic structure **350** may be maintained in the basic state, and a partial area of the connecting member **340** may be supported while contacting the rod of the elastic structure **350**, the elastic structure **350** may be bent into a shape, in which it is located between the first protruding part **334** of the second rear case **332** and the elastic structure **350**. According to an example embodiment, when an external force is applied to the electronic device **300** and then is removed, the connecting member **340** may be automatically restored to the bent shape before the external force is applied through an operation of the elastic structure **350**. The operation of the elastic structure **350** and the movement of the connecting member **340** will be described in more detail with reference to FIG. **12**.

[0139] The rear plate **380** may define a rear surface of the electronic device **300**. The rear plate **380** may be coupled to the side member **310** to cover the battery **391** and the rear case (e.g., the second rear case **332** and the first rear case **331** of FIG. **4A**).

[0140] FIG. **10** is a view illustrating a rear case and a connecting member of an electronic device according to an embodiment. FIG. **11** is a cross-sectional view of an electronic device according to an embodiment.

[0141] FIG. **10** illustrates a plan view of a portion of an electronic device, in which the connecting member and the rear case are disposed, and a view illustrating a shape of the rear case. FIG. **11** is a view illustrating cross section D-D' of the electronic device according to an embodiment illustrated in FIG. **10**.

[0142] Referring to FIGS. **10** and **11**, an electronic device **300** according to an embodiment may include a side member **310**, a second circuit board **322**, a second rear case **332'**, a connecting member **340**, an elastic structure **350**, a rear plate **380**, and a battery **391**.

[0143] The embodiment illustrated in FIGS. **10** and **11** relates to a second rear case **332'** having a shape that is different from that of the second rear case **332** according to the embodiments of FIGS. **8**, **9A**, and **9B**, and hereinafter, a repeated description thereof will be omitted, and their differences will be mainly described.

[0144] Unlike in the second rear case **332** according to the first embodiment illustrated in FIGS. **8**, **9A**, and **9B**, in the second rear case **332'** according to the second embodiment illustrated in FIGS. **10** and **11**, cutaway portions (e.g., the cutaway portions **336** and **337** of FIG. **8**) may not be formed in the first protruding part **334'** and the second protruding part **335'**.

[0145] The second rear case **332'** may be formed in a shape, in which the first protruding part **334'** and the second protruding part **335'** continuously extend in a width "W" direction of the connecting member **340** in correspondence to the width "W" of the connecting member **340**. For example, the first protruding part **334'** may continuously cover a partial area of the connecting member **340** disposed under the first protruding part **334'**. For example, the second protruding part **335'** may continuously support a partial area of the connecting member **340** that extends into a space between the second protruding part **335'** and the plate part **312**. The second rear case **332'** according to the second embodiment may be manufactured through computer numerical control (CNC) processing, but the disclosure is not limited thereto.

[0146] Compared to the second rear case **332** according to the first embodiment, the second rear case **332'** according to the second embodiment may more effectively provide a function of preventing or reducing the connecting member **340** from pushing out the rear plate **380** or colliding with the rear plate **380** through the first protruding part **334'** because an area of the first protruding

part **334'**, which overlaps the connecting member **340**, is relatively large. Compared to the second rear case **332** according to the first embodiment, the second rear case **332'** according to the second embodiment may more effectively provide a function of formalizing a bent shape of a portion of the connecting member **340** and alleviating or controlling rapid movement of the connecting member **340** through the second protruding part **335'** because an area of the second protruding part **335'**, which supports the connecting member **340**, is relatively large.

[0147] FIG. **12** is a view illustrating driving of an elastic structure and movement of a connecting member of an electronic device according to an embodiment.

[0148] Referring to FIG. **12**, an electronic device **300** according to an embodiment may include a side member **310** (e.g., the plate part **312**), a second circuit board **322**, a second rear case **332**, a connecting member **340**, an elastic structure **350**, and a rear plate **380**.

[0149] At least some of the components of the electronic device **300** illustrated in FIG. **12** may be substantially the same as or similar to the components of the electronic device **300** illustrated in FIGS. **8** to **11**, and hereinafter, repeated descriptions thereof will be omitted.

[0150] **1201** of FIG. **12** illustrates an operation of maintaining the elastic structure **350** in a basic state as an external force or an impact is not applied to the electronic device **300** or the applied external force or impact is removed. **1203** of FIG. **12** illustrates an operation of deforming the elastic structure **350** from the basic state as an external force or an impact that is directed in the first direction {circle around (1)} is applied to the electronic device **300**. For example, when an external force is applied to the electronic device in the first direction {circle around (1)} in a state illustrated in **1201**, the electronic device **300** may be deformed into a state of **1203**, and when an external force in the first direction {circle around (1)} applied to the electronic device **300** is removed, or an external force is applied in the second direction {circle around (2)}, the electronic device **300** may be deformed into a state of **1201**.

[0151] When an external force or an impact is applied to the electronic device **300** in the state illustrated in **1201** in the first direction {circle around (1)} in an operation of deforming the electronic device **300** from the state of **1201** to the state of **1203**, at least a portion of the connecting member **340** may be moved in the first direction {circle around (1)}. As the connecting member **340** is moved in the first direction {circle around (1)}, an external force may be applied to the rod **370** of the elastic structure **350** in the first direction {circle around (1)}, and the rod **370** may be rotated within a specific range in the first rotational direction (e.g., a direction corresponding to the first direction {circle around (1)}). The movement of the connecting member **340** may be guided or controlled by a rotation operation of the rod **370**, and an impact applied to the connecting member **340** may be minimized or reduced. The rod **370** may be rotated in the first rotational direction (e.g., a direction that becomes closer to a side wall **314** of the plate part **312**), and a maximum or large rotation degree thereof may be restricted by the side wall **314**. For example, the rod **370** may be configured not to be rotated in the first rotational direction any more when contacting the side wall **314**. When being moved in the first direction {circle around (1)}, the connecting member **340** may not contact the rear plate **380** as it contacts the inclined surface **334s** of the first protruding part **334** of the second rear case **332** or is moved along the inclined surface **334s**.

[0152] When the electronic device **300** is deformed from the state of **1201** to the state of **1203** by an external force that is directed in the first direction {circle around (1)}, the rod **370** of the elastic structure **350** may be rotated in the second rotational direction that is opposite to the first rotational direction by a repulsive force (e.g., a repulsive force that occurs due to the twisting of the rod **370**) and may return to a position of the basic state when the external force is removed or an external force that is directed in the second direction {circle around (2)} is applied. At least a portion of the connecting member **340** may be moved in the second direction {circle around (2)} as the rod **370** is rotated, and may be restored to the state of **1201**. The movement of the connecting member **340** may be guided or controlled by a rotation operation of the rod **370**, and an impact applied to the connecting member **340** may be minimized or reduced. When being moved in the second direction

{circle around (2)}, the connecting member **340** may not contact the rear plate **380** as it contacts the inclined surface **334s** of the first protruding part **334** of the second rear case **332** or is moved along the inclined surface **334s**. The bent shape of the connecting member **340** may be automatically formalized into the shape illustrated in **1201** as the elastic structure **350** is restored to the basic state through the repulsive force of the rod **370**.

[0153] According to various embodiments, when an excessive external force is applied to the electronic device **300** in the second direction {circle around (2)}, the rod **370** may be rotated in the second rotational direction, and the maximum or large rotation degree thereof may be restricted by the second protruding part **335**. For example, the rod **370** may be configured not to be rotated in the second rotational direction any more when contacting the second protruding part **335**.

[0154] FIG. **13** is a view illustrating a method for assembling an elastic structure, a second circuit board, a connecting member, and a second rear case of an electronic device according to an embodiment.

[0155] Referring to FIG. **13**, the assembly method may include an operation **1301** of assembling an elastic structure **350**, an operation **1303** of assembling a second circuit board **322**, an operation **1305** of assembling a connecting member **340**, and an operation **1307** of assembling a second rear case **332**. For example, the operation **1301** of assembling the elastic structure **350**, the operation **1303** of assembling the second circuit board **322**, the operation **1305** of assembling the connecting member **340**, and the operation **1307** of assembling the second rear case **332** may be sequentially performed.

[0156] In the operation **1301** of assembling the elastic structure **350**, the elastic structure **350** may be attached to the plate part **312** of the side member **310** through an adhesive member **393**. The elastic structure **350** may be coupled to a partial area of the plate part **312** that is adjacent to the battery **391**. According to various embodiments, the assembly of the elastic structure **350** through the adhesive member **393** may be omitted. A through-hole **315**, through which the connecting member **340** passes, may be formed in the plate part **312**, and the through-hole **315** may be partially aligned with the opening **351** of the elastic structure **350**.

[0157] In the operation **1303** of assembling the second circuit board **322**, the second circuit board **322** may be coupled to the plate part **312** through the first fixing member **392**. The second circuit board **322** may be disposed in the plate part **312** to overlap the holder **360** of the elastic structure **350**, which is attached to the plate part **312**, and the first fixing member **392** may pass through the second circuit board **322** and a portion of the holder **360** to be fastened to the plate part **312**. For example, the elastic structure **350** may be coupled to the plate part **312** primarily through the adhesive member **393** and secondarily through the first fixing member **392**. When the assembly of the second circuit board **322** is completed, the rod **370** of the elastic structure **350** may be located between the sidewall **314** of the plate part **312** and the circuit board **320**.

[0158] In operation **1305** of assembling the connecting member **340**, the connecting member **340** may be assembled to pass through the opening **351** of the elastic structure **350** and the through-hole **315** of the plate part **312**. The connecting member **340** extends across an upper side of the battery **391**, and may pass through the opening **351** and the through-hole **315**.

[0159] In the operation **1307** of assembling the second rear case **332**, the second rear case **332** may be coupled to the plate part **312** to cover the second circuit board **322**. For example, the second rear case **332** may be coupled to the plate part **312** through the second fixing member **394**, and the second fixing member **394** may pass through the second rear case **332** and the second circuit board **322** to be fastened to the plate part **312**.

[0160] Hereinafter, a structure of the elastic structure according to various embodiments disclosed in the disclosure will be described with reference to FIGS. **14** to **19**. For example, the elastic structures described with reference to FIGS. **14** to **19** may have different shapes and/or structures from those of the elastic structures according to the above-described embodiments (e.g., the elastic structure **350** of FIGS. **5** to **7**), but may have substantially the same purpose and/or function and

may have substantially the same overall operation. Hereinafter, in the description of the embodiments of FIGS. **14** to **19**, the repeated contents of the above description will be omitted, and differences thereof will be mainly described.

[0161] FIG. **14** is a view illustrating a connecting member and an elastic structure of an electronic device according to an embodiment. FIG. **15** is a view illustrating an elastic structure of an electronic device according to an embodiment.

[0162] Referring to FIGS. **14** and **15**, an elastic structure **400** according to an embodiment may include a holder **410** (e.g., the holder **360** of FIGS. **5** to **7**) and a rod **430** (e.g., the rod **370** of FIGS. **5** to **7**) that is coupled to the holder **410**. The elastic structure **400** may be formed in a form including a fixing shaft to further increase elasticity (or tension). For example, the elastic structure **400** according to the embodiment illustrated in FIGS. **14** and **15** may be a structure, in which a portion (e.g., a fixing part **434**) that defines the fixing shaft is added to the rod and a portion (e.g., a third coupling part **414**), to which the fixing part **434** is coupled, is added to the holder **360** in the elastic structure **350** of FIGS. **5** to **7**. Similar to the elastic structure **350** of FIGS. **5** to **7**, the elastic structure **400** of FIGS. **14** and **15** may provide elasticity by using a repulsive force that is generated while a portion and another portion of the rod **430** are rotated around the rotation axes R1 and R2 located at different heights, respectively.

[0163] The holder **410** may include a first base part **411** (e.g., the first base part **364** of FIG. **6**), a first coupling part **412** (e.g., the first coupling part **365** of FIG. **6**) that extends from one end of the first base part **411**, a second coupling part **413** (e.g., the second coupling part **366** of FIG. **6**) that extends from an opposite end of the first base part **411**, and a third coupling part **414** that extends from the first coupling part **412**. The third coupling part **414** may extend from the first coupling part perpendicularly thereto.

[0164] The rod **430** may include a second base part **431** (e.g., the second base part **374** of FIG. **5**), a first extension part **432** (e.g., the first extension part **375** of FIG. **5**) that extends from one end of the second base part **431**, a second extension part **433** (e.g., the second extension part **376** of FIG. **6**) that extends from an opposite end of the second base part **431**, and a fixing part **434** that extends from the first extension part **432**. The first extension part **432** may be coupled to the first coupling part **412** of the holder **410** to be rotatable. The second extension part **433** may be coupled to the second coupling part **413** of the holder **410** to be rotatable. The fixing part **434** may be fixedly coupled to the third coupling part **414** of the holder **410**. The fixing part **434** may be parallel to the first extension part **432** and the second extension part **433**. The fixing part **434** and the first extension part **432** may be connected to each other by a connecting part **435**. For example, the connecting part **435** may be perpendicular to the first extension part **432** and the fixing part **434**. The fixing part **434** and the first extension part **432** may overlap each other with respect to a direction, in which the connecting part **435** extends.

[0165] The first extension part **432** may form or define the first rotation axis R1. The second extension part **433** may form or define the second rotation axis R2. The first rotation axis R1 may be located closer to the plate part **312** than the second rotation axis R2. For example, the rod **430** may be formed in a shape, in which the first extension part **432** and the second extension part **433** are spaced apart from a portion of the first base part **411** that is parallel to the rotation axes R1 and R2 by different distances.

[0166] The fixing part **434** may form or define a fixing shaft. The fixing shaft indicated by the fixing part **434** may be parallel to the first rotation axis R1 and the second rotation axis R2, but may be located at a different height from the plate part **312**. For example, the fixing part **434** may be located more distant from the plate part **312** than the first extension part **432** and the second extension part **433**. The fixing part **434** may be a reference for a torsional displacement that occurs in the remaining portion as it is fixedly coupled to the third coupling part **414** of the holder **410**. For example, the rod **430** may increase the repulsive force (e.g., an elastic force) generated by torsion as the fixing part **434** is fixed without rotating together with the remaining portions of the

part.

[0167] FIG. **16** is a view illustrating a connecting member and an elastic structure of an electronic device according to an embodiment. FIG. **17** is a view illustrating an elastic structure of an electronic device according to an embodiment.

[0168] Referring to FIGS. **16** and **17**, an elastic structure **500** according to an embodiment may be formed in the form of a rod that is directly coupled to the side member **310** without a holder. For example, the elastic structure **500** according to an embodiment illustrated in FIGS. **16** and **17** may be a modified structure, in which the holder (e.g., the holder **410** of FIGS. **14** and **15**) is omitted from the elastic structure (e.g., the elastic structure **400** of FIGS. **14** and **15**) of FIGS. **14** and **15** and the rod (e.g., the rod **430** of FIGS. **14** and **15**) is directly coupled to the side member **310**.

[0169] The elastic structure **500** may be formed by partially bending a bar having a specific rigidity. The elastic structure **500** may include a base part **510**, a first extension part **520** that extends from one end of the base part **510**, a second extension part **530** that extends from an opposite end of the base part **510**, and a fixing part **540** that extends from the first extension part **520**. The first extension part **520**, the second extension part **530**, and the fixing part **540** may be parallel to each other. The fixing part **540** and the first extension part **520** may be connected to each other by a connecting part **550**. For example, the connecting part **550** may be perpendicular to the first extension part **520** and the fixing part **540**. The fixing part **540** and the first extension part **520** may overlap each other with respect to a direction, in which the connecting part **550** extends.

[0170] The first extension part **520** may form or define the first rotation axis R1. The second extension part **530** may be coupled to the side member **310** to be rotatable, and may form or define the second rotation axis R2. The first rotation axis R1 may be located closer to the plate part **312** than the second rotation axis R2. For example, the elastic structure **500** may be formed in a shape, in which the first extension part **520** and the second extension part **530** are spaced apart from a portion of the base part **510** that is parallel to the rotation axes R1 and R2 by different distances.

[0171] The fixing part **540** may be fixed to the side member **310**. For example, the fixing part **540** may form or define a fixing shaft. The fixing shaft indicated by the fixing part **540** may be parallel to the first rotation axis R1 and the second rotation axis R2, but may be located at a different height from the plate part **312**. For example, the fixing part **540** may be located more distant from the plate part **312** than the first extension part **520** and the second extension part **530**. The fixing part **540** may be a reference for a torsional displacement that occurs in the remaining portion as it is fixedly coupled to the plate part **312**. For example, the elastic structure **500** may increase the repulsive force (e.g., an elastic force) generated by torsion as the fixing part **540** is fixed without rotating together with the remaining portions of the part.

[0172] When an external force is applied to the elastic structure **500** in the first direction {circle around (1)}, the elastic structure **500** may be partially twisted while the first extension part **520** and the second extension part **530** of the elastic structure **500** are rotated in the first rotational direction (e.g., a direction that becomes closer to the fixing part **540**) around the first rotation axis R1 and the second rotation axis R2, respectively. When the external force is removed, the first extension part **520** and the second extension part **530** may return to the state before the external force is applied while being rotated in an opposite direction (e.g., a direction that becomes more distant from the fixing part **540**) to the first rotation direction around the first rotation axis R1 and the second rotation axis R2, respectively, by the restoring force of the elastic structure **500** in a twisted state.

[0173] FIG. **18** is a view illustrating a connecting member and an elastic structure of an electronic device according to an embodiment. FIG. **19** is a view illustrating an elastic structure of an electronic device according to an embodiment.

[0174] Referring to FIGS. **18** and **19**, in an elastic structure **600** according to an embodiment, a rod **630** that supports the connecting member **340** may be integrally formed with a holder **610**. Unlike the elastic structures **350**, **400**, and **500** of the previous embodiments, the elastic structure **600** according to an embodiment illustrated in FIGS. **18** and **19** may provide a specific tension even

when the rotation axes “R” are not located at different heights as the holder **610** itself is formed of a material having elasticity.

[0175] The holder **610** of the elastic structure **600** may be formed of a material having elasticity. The holder **610** may include a first portion **611**, and second portions **613** that extend from opposite ends of the first portion **611**. A rod **630** of the elastic structure **600** may be connected to the second portion **613** of the holder **610**. The holder **610** may be moved in a direction, in which the second portion **613** becomes closer to or more distant from the first portion **611** by an external force, and when the external force is removed, the holder **610** may return to a state before the external force is applied by the elastic force (or a restoring force) of the holder **610**. The holder **610** is a thin plate-shaped spring (e.g., a leaf spring), and may have an elastic force of a specified magnitude. For example, the holder **610** may be formed of a plate spring made of a metal material, but the disclosure is not limited thereto.

[0176] When an external force is applied to the rod **630** in the first direction {circle around (1)}, the rod **630** may be rotated around the rotation axis “R” together with the second portion **613** in the first rotational direction (e.g., a direction that becomes more distant from the first portion **611**) with respect to the first portion **611**, and when the external force is removed, the rod **630** may be rotated around the rotation axis “R” together with the second portion **613** in an opposite direction (e.g., a direction that becomes closer to the first portion **611**) to the first rotational direction, by the elastic force of the holder **610**, to return to its original position.

[0177] The electronic device may include a connecting member (e.g., an FPCB which of course include conductive material and dielectric material) for connecting an upper substrate and a lower substrate. The connecting member may extend by a sufficient length to prevent or reduce separation from the substrate. As the connecting member is formed long, it may be partially moved in an interior of the electronic device due to an impact or an external force applied to the electronic device. As the connecting member is moved, the connecting member may collide with the rear cover of the electronic device, and thus, lifting may occur in the rear cover.

[0178] Certain example embodiments provide an electronic device, in which movement of the connecting member may be guided through the elastic structure, and a bent shape of which may be formalized.

[0179] An electronic device **300** according to an example embodiment may include a housing **301** including a plate part **312**, a first circuit board **321** that is disposed in the plate part to be adjacent to one side of the housing, which faces the first direction {circle around (1)}, a second circuit board **322** that is disposed in the plate part to be adjacent to an opposite side of the housing, which faces the second direction {circle around (2)} that is opposite to the first direction, a rear case **332** and **332'** that is coupled to the plate part to cover at least a portion of the second circuit board, an elastic structure **350**, **400**, **500**, and **600** coupled to the plate part to be adjacent to the second circuit board, and configured to provide an elastic force, and a connecting member **340** connecting the first circuit board and the second circuit board, and a portion of which contacts the elastic structure. The rear case may include a first protruding part **334** and **334'** that faces a portion of the elastic structure and a portion of the connecting member, which contacts the elastic structure.

[0180] In various embodiments, the connecting member may be movable in the housing, and movement of the connecting member may be guided by the elastic structure or the first protruding part.

[0181] In various embodiments, the elastic structure may be configured such that at least a portion thereof is rotated toward the first direction or the second direction in correspondence to an external force applied to the electronic device.

[0182] The connecting member may be configured such that at least a portion thereof is moved in the first direction or the second direction based on rotation of the elastic structure.

[0183] In various embodiment, the elastic structure may include a holder **360**, **410**, and **610** coupled to the plate part, and a rod **370**, **430**, and **630** coupled to the holder. The rod may be

configured such that a first end thereof is coupled to the holder to be rotatable around a first rotation axis **R1**, and a second end thereof being opposite to the first end is coupled to the holder to be rotatable around a second rotation axis **R2**. The first rotation axis and the second rotation axis may be parallel to each other, and may be spaced apart from each other by a specific distance.

[0184] In various embodiments, the elastic structure may be configured such that the first rotation axis is located at a first height **H1** from the holder in a third direction {circle around (3)}, and the second rotation axis is located at a second height **H2** from the holder in the third direction. The third direction may be a direction being perpendicular to the first direction and the second direction, and facing a rear surface of the electronic device.

[0185] In various embodiments, the first rotation axis and the second rotation axis may be arranged in a direction being parallel to the third direction.

[0186] In various embodiments, the elastic structure may be configured such that an opening **351** is formed between the rod and the holder. The connecting member may contact the rod and may extend to pass through the opening.

[0187] In various embodiment, a through-hole **315** aligned with the opening may be formed in the plate part. A portion of the connecting member may extend, one surface of the plate part, which faces the third direction, to an opposite surface of the plate part, which faces an opposite direction to the third direction across the rod, the opening, and the through-hole.

[0188] In various embodiments, the first protruding part may be spaced apart from the rod of the elastic structure, and a surface thereof, which faces the connecting member, may be formed to be an inclined surface **334s**.

[0189] In various embodiment, the rear case further may include a body part **333** and **333'** coupled to the plate part. The first protruding part may protrude from the body part in the first direction.

[0190] In various embodiments, the rear case may further include a second protruding part **335** and **335'** protruding from the body part in the first direction such that at least a portion thereof is located in an opening of the elastic structure.

[0191] In various embodiments, at least a portion of the connecting member may be disposed in a space between a portion of the plate part and the second protruding part, and is bent in a specific shape in correspondence to a shape of the second protruding part.

[0192] In various embodiments, the rod and a partial area of the connecting member, which contacts the rod, may overlap the first protruding part and the second protruding part in the third direction.

[0193] In various embodiments, the holder may include a first base part **364** and **411** fixed to the plate part; a first coupling part **365** and **412** extending from one end of the first base part; and a second coupling part **366** and **413** extending from an opposite end of the first base part, and the first coupling part and the second coupling part may extend to be located at different heights from the first base part.

[0194] In various embodiments, the rod may include a second base part **374** and **431** which the connecting member contacts; a first extension part **375** and **432** extending from one end of the second base part and coupled to the first coupling part to be rotatable; and a second extension part **376** and **433** extending from an opposite end of the second base part and coupled to the second coupling part to be rotatable.

[0195] In various embodiments, the holder may further include a third coupling part **414** extending from the first coupling part. The rod may further include a fixing part **434** extending from the first extending part, and fixedly coupled to the third coupling part.

[0196] An electronic device **300** according to an example embodiment may include a housing **301** including a side member **310**, wherein the side member may include a plate part **312** and a frame part **311** surrounding a periphery of the plate part, a first circuit board **321** disposed in the plate part adjacent to one side of the housing, which faces a first direction {circle around (1)}, a second circuit board **322** disposed in the plate part adjacent to an opposite side of the housing, which faces

a second direction {circle around (2)} being opposite to the first direction, a rear case **332** and **332'** coupled to the plate part to cover at least a portion of the second circuit board, an elastic structure **350**, **400**, **500**, and **600** coupled to the plate part adjacent to the second circuit board, and configured to provide an elastic force, and a connecting member **340** connecting the first circuit board and the second circuit board, and a portion of which contacts the elastic structure. Movement of the connecting member may be guided in an interior of the housing based on elastic deformation of the elastic structure. "Connecting" as used herein covers both direct and indirect connections.

[0197] In various embodiments, at least a portion of the elastic structure may be rotated toward the first direction or the second direction as an external force is applied to the elastic structure in a basic state, and may return to the basic state as the external force is removed.

[0198] In various embodiment, the elastic structure may include a holder **360**, **410**, and **610** coupled to the plate part, and a rod **370**, **430**, and **630** coupled to the holder. The rod may be configured such that a first end thereof is coupled to the holder to be rotatable around a first rotation axis R1, and a second end thereof being opposite to the first end is coupled to the holder to be rotatable around a second rotation axis R2. The first rotation axis and the second rotation axis may be parallel to each other, and may be spaced apart from each other by a specific distance.

[0199] In various embodiments, the elastic structure may be configured such that the first rotation axis is located at a first height H1 from the holder in a third direction {circle around (3)}, and the second rotation axis is located at a second height H2 from the holder in the third direction. The third direction may be a direction being perpendicular to the first direction and the second direction, and facing a rear surface of the electronic device.

[0200] The electronic device according to certain example embodiments may prevent or reduce chances of an impact of the connecting member by optimizing and controlling movement or deformation of the connecting member through the elastic structure.

[0201] Besides, a variety of effects directly or indirectly understood through the disclosure may be provided.

[0202] The electronic device according to certain embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0203] It should be appreciated that certain embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as "A or B," "at least one of A and B," "at least one of A or B," "A, B, or C," "at least one of A, B, and C," and "at least one of A, B, or C," may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as "1.sup.st" and "2.sup.nd," or "first" and "second" may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term "operatively" or "communicatively", as "coupled with," "coupled to," "connected with," or "connected to" another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element. Thus, "coupled" and "connecting" as used herein cover both direct and indirect coupling, and direct and indirect connections.

[0204] As used in connection with various embodiments of the disclosure, the term "module" may include a unit implemented in hardware, software, or firmware, and may interchangeably be used

with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC). Thus, each “module” herein may comprise circuitry.

[0205] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0206] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0207] According to certain embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities and some of multiple entities may be separately disposed on the other components. According to certain embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to certain embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to certain embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

1. An electronic device comprising: a housing including a plate part; a first circuit board disposed in and/or supported by the plate part adjacent to a side of the housing, which faces a first direction; a second circuit board disposed in and/or supported by the plate part adjacent to an opposite side of the housing which faces a second direction being opposite to the first direction; a rear case coupled to the plate part to cover at least a portion of the second circuit board; an elastic structure, comprising elastic material, coupled to the plate part adjacent to the second circuit board, and configured to provide an elastic force; and a connecting member, comprising conductive material, electrically connecting the first circuit board and the second circuit board, and a portion of which

contacts the elastic structure, wherein the rear case includes a first protruding part, comprising a protrusion, facing a portion of the elastic structure and a portion of the connecting member, which contacts the elastic structure.

2. The electronic device of claim 1, wherein the connecting member is configured to be movable in the housing, and wherein the connecting member is configured so that movement of the connecting member is guided by the elastic structure and/or the first protruding part.

3. The electronic device of claim 1, wherein the elastic structure is configured such that at least a portion thereof is configured to be rotated toward the first direction or the second direction in correspondence to an external force applied to the electronic device, and wherein the connecting member is configured such that at least a portion thereof is configured to be moved in the first direction or the second direction based on rotation of the elastic structure.

4. The electronic device of claim 1, wherein the elastic structure includes: a holder coupled to the plate part; and a rod coupled to the holder, wherein the rod is configured such that a first end thereof is coupled to the holder and configured to be rotatable around a first rotation axis, and a second end thereof being opposite to the first end is coupled to the holder and configured to be rotatable around a second rotation axis, and wherein the first rotation axis and the second rotation axis are substantially parallel to each other, and are spaced apart from each other.

5. The electronic device of claim 4, wherein the elastic structure is configured such that the first rotation axis is located at a first height from the holder in a third direction, and the second rotation axis is located at a second height from the holder in the third direction, and wherein the third direction is a direction being perpendicular to the first direction and the second direction, and facing a rear surface of the electronic device.

6. The electronic device of claim 5, wherein the first rotation axis and the second rotation axis are arranged in a direction being substantially parallel to the third direction.

7. The electronic device of claim 5, wherein the elastic structure is configured such that an opening is formed between at least the rod and the holder, and wherein the connecting member contacts the rod and extends to pass through the opening.

8. The electronic device of claim 7, wherein a through-hole aligned with the opening is formed in the plate part, and wherein a portion of the connecting member extends, on one surface of the plate part, which faces the third direction, to an opposite surface of the plate part, which faces an opposite direction to the third direction across the rod, the opening, and the through-hole.

9. The electronic device of claim of 8, wherein the first protruding part is spaced apart from the rod of the elastic structure, and a surface thereof, which faces the connecting member, is formed to be an inclined surface.

10. The electronic device of claim 1, wherein the rear case further includes a body part coupled to the plate part, and wherein the first protruding part protrudes from the body part in the first direction.

11. The electronic device of claim 10, wherein the rear case further includes: a second protruding part, comprising a protrusion, protruding from the body part in the first direction such that at least a portion thereof is located in an opening of the elastic structure.

12. The electronic device of claim 11, wherein at least a portion of the connecting member is disposed in a space between a portion of the plate part and the second protruding part, and is bent in a specific shape in correspondence to a shape of the second protruding part.

13. The electronic device of claim 11, wherein the rod and a partial area of the connecting member, which contacts the rod, overlap the first protruding part and the second protruding part in the third direction.

14. The electronic device of claim 4, wherein the holder includes: a first base part fixed to the plate part; a first coupling part extending from one end of the first base part; and a second coupling part extending from an opposite end of the first base part, and wherein the first coupling part and the second coupling part extend to be located at different heights from the first base part.

- 15.** The electronic device of claim 14, wherein the rod includes: a second base part which the connecting member contacts; a first extension part extending from one end of the second base part and coupled to the first coupling part to be rotatable; and a second extension part extending from an opposite end of the second base part and coupled to the second coupling part to be rotatable.
- 16.** The electronic device of claim 15, wherein the holder further includes a third coupling part extending from the first coupling part, and wherein the rod further includes a fixing part extending from the first extending part, and fixedly coupled to the third coupling part.
- 17.** An electronic device comprising: a housing including a side member, wherein the side member includes a plate part and a frame part surrounding a periphery of the plate part; a first circuit board disposed in and/or supported by the plate part adjacent to a side of the housing, which faces a first direction; a second circuit board disposed in and/or supported by the plate part adjacent to an opposite side of the housing, which faces a second direction opposite to the first direction; a rear case coupled to the plate part to cover at least a portion of the second circuit board; an elastic structure, comprising elastic material, coupled to the plate part adjacent to the second circuit board, and configured to provide an elastic force; and a connecting member, comprising a dielectric material and/or a conductive material, connecting the first circuit board and the second circuit board, and a portion of which contacts the elastic structure, wherein the connecting member and the elastic structure are configured so that movement of the connecting member is guided in an interior of the housing based on elastic deformation of the elastic structure.
- 18.** The electronic device of claim 17, wherein at least a portion of the elastic structure is configured to be rotated toward the first direction or the second direction as an external force is applied to the elastic structure in a basic state, and returns to the basic state as the external force is removed.
- 19.** The electronic device of claim 17, wherein the elastic structure includes: a holder coupled to the plate part; and a rod coupled to the holder, wherein the rod is configured such that a first end thereof is coupled to the holder to be rotatable around a first rotation axis, and a second end thereof being opposite to the first end is coupled to the holder to be rotatable around a second rotation axis, and wherein the first rotation axis and the second rotation axis are parallel to each other, and are spaced apart from each other.
- 20.** The electronic device of claim 19, wherein the elastic structure is configured such that the first rotation axis is located at a first height from the holder in a third direction, and the second rotation axis is located at a second height from the holder in the third direction, and wherein the third direction is a direction being perpendicular to the first direction and the second direction, and facing a rear surface of the electronic device.
-